Problem 3.4

Assume that $A(t) = 225 - (t - 10)^2$, $0 \leq t \leq 10$. Find i_6.

Solution

By definition,

$$i_6 = \frac{A(6) - A(5)}{A(5)}.$$

Now compute the required values:

$$A(6) = 225 - (6 - 10)^2 = 225 - 16 = 209,$$

and

$$A(5) = 225 - (5 - 10)^2 = 225 - 25 = 200.$$

Substituting,

$$i_6 = \frac{209 - 200}{200}.$$

Thus,

$$i_6 = \frac{9}{200} = 0.045.$$

Therefore,

$$\boxed{i_6 = 4.5\%.}$$